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Skill-Targeted Universal Social and Emotional Learning Interventions for Education in Conflict and Crisis: Experimental Impacts in Niger

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ABSTRACT

Universal, school-based social-emotional learning (SEL) interventions hold the promise to promote children's academic learning and social and emotional skills necessary to cope with adversity in the context of education in conflict and crisis. This study evaluated the sequential implementation of two sets of skill-targeted SEL activities—first, testing the impact of 11-week implementation of Mindfulness activities alone, then the combined effect of Mindfulness and Brain Games implemented sequentially over the course of 22 weeks—with Nigerian refugee and Nigerien local second to fourth graders (N = 1,795) attending a remedial education programming in Diffa, Niger, via a cluster randomized controlled trial. Positive impacts of Mindfulness activities were observed on a select number of children's SEL outcomes, with some additional benefits for refugees and girls, demonstrating the potential for low-cost SEL approaches to support vulnerable children at scale in crisis-affected contexts. These findings provide actionable insights for policymakers, program developers and implementers in conflict-affected, low-income countries, such as the potential for targeted interventions to improve SEL outcomes for marginalized children and the need to invest in contextually-responsive and sustained program models to maximize long-term impact.

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Since the mid-2010s, violent attacks by Boko Haram—a jihadist militant group whose name translates into "Western education is forbidden"—have led over 240,000 Nigerian refugees and internally-displaced Nigeriens to seek protection in Diffa (UNHCR, 2021), a southeast region of Niger bordering Northern Nigeria with a total population estimate of 815,326 (OCHA, 2022). Boko Haram's violence and atrocities targeting education staff, students, and spaces have resulted in severe disruptions to Nigerian and Nigerien children's education in the region, often affecting their academic and social-emotional development that was already threatened by poverty, malnutrition,

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and hazardous environments (Save the Children, 2021; UNICEF, 2021). Extant research suggests that children affected by conflict and violence have greater difficulty developing positive social, emotional, and cognitive skills that can help them navigate the adversity they face in the conflict- and crisis-affected contexts (Boxer et al., 2013; Buckner & Kim, 2012; Cummings, Merrilees, et al., 2010; Cummings, Schermerhorn, et al., 2010; Dubow et al., 2019; Khamis, 2019; Qouta et al., 2008; Song et al., 2023).

Universal, school-based social-emotional learning (SEL) interventions hold the promise of proactively supporting social and emotional development necessary to help cope with exposure to adversity in conflict-affected contexts (Betancourt et al., 2013). Although universal school-based SEL programs have been designed, implemented, and rigorously evaluated in the U.S. and other high-income countries for nearly 50 years with demonstrated impact on improving not only social and emotional skills but also academic and life outcomes in a long term (Cipriano et al., 2023; Durlak et al., 2011), only in the last decade have they begun to be adapted, used, and evaluated in conflict- and crisis-affected low-income countries. The extant evidence of effectiveness in such settings remains scant and inconclusive, often reporting small or null impacts (Deitz et al., 2021; Kim et al., 2024). In addition, the extant evaluations focus mostly on psychological distress and mental health outcomes (e.g., trauma symptoms), and often do not examine the effects on the actual SEL skills that the interventions target, which is inconsistent with underlying frameworks and theory of change of the SEL interventions (Lasater et al., 2022). Evidence is even more limited on whether universal SEL interventions can provide much-needed support for marginalized subgroups of children, such as girls and refugees, when such interventions are not directly or exclusively targeting such a population (Kim et al., 2024). In this study, we provide some of the first experimental evidence of a school-based, universal SEL approach designed for low-resource conflict-affected contexts: *skill-targeted SEL activities* layered on top of an existing tutoring program (Tutoring in Healing Classrooms (HCT)). Specifically, this study reports the average and differential impacts of the sequential implementation of two sets of skill-targeted SEL activities—Mindfulness activities and Brain Games—embedded within the HCT framework in the Boko Haram-affected Diffa region of Niger. By demonstrating the feasibility and impact of these low-cost interventions, this study offers actionable insights for policymakers, program developers and implementers seeking scalable strategies to enhance educational programming in crisis-affected settings. Specifically, it highlights ways to support children's social-emotional development while addressing equity gaps for marginalized populations, including refugees and girls, in challenging environments.

Context of Conflict and Crisis: Children's Experience in Diffa

Niger is ranked the "toughest place to be a child" (Save the Children, 2021, p.1), with children facing threats to education, health, child labor, malnutrition, and violence. Located in the Lake Chad Basin and bordering Northern Nigeria, Niger's Diffa region has been a temporary home to nearly 130,000 Nigerian refugees and over 110,000 Nigerien internally displaced persons (IDPs) (UNHCR, 2021), with about half of them being school-aged children (UNHCR, 2020). The majority (87%) of these displaced people lived in informal settlements, often within or near host communities (UNHCR, 2021). More than 15% of the population in Diffa (including Fulani, Tuareg, and Diffa Arabs ethnic groups) lead a traditional nomadic lifestyle, and their seasonal migration affects the continuity of children's education. Frequent attacks, suicide bombings, and kidnapping by Boko Haram—as well as the severe counter-insurgency measures and surveillance on locals—have terrorized the Diffa, exacerbating local intercommunal tensions and engendering an atmosphere of collective fear (International Crisis Group, 2017).

Marginalized groups, including Nigerian refugee children and girls, are more vulnerable to these challenges. Though public schools are open to all children, Nigerian refugee children experience additional challenges in schools due to the disruption in their education as a result of displacement. They also struggle to engage in learning activities in classrooms due to the lack of proficiency in French—the official language of instruction in Niger. On the other hand, girls in Niger face child marriage, early and frequent childbirth, and unpaid domestic work (Save the Children, 2021). As a result, many Nigerien girls drop out of school early, and even if they stay, have little time to go to school and engage in schoolwork. Existing evidence on SEL programming in conflict-affected contexts suggests that the benefits of SEL programs may vary by gender and refugee status, but this variability is likely dependent on the context, population, and program features (Kim et al., 2024). Given the heightened adversities these groups face, it is critical to understand how SEL interventions impact their SEL and whether these interventions can help close equity gaps.

Universal SEL for Education in Conflict and Crisis

Whereas the protection and health of the children should be prioritized for those directly affected by violence and severe adversity, children affected by conflict and crisis can also benefit from support that bolsters skills they can use to navigate challenging conditions. Universal SEL programs, designed to promote social and emotional skill development as a part of the learning process, can serve as one such solution to support resilience in crisis contexts, proactively addressing risks and building competencies at scale (Aber et al., 2021; Jordans et al., 2016).

Systematic reviews suggest that school-based universal approaches to SEL interventions, offered in education settings to all children regardless of vulnerability, provide the most potential to cost-effectively promote healthy development and learning for all (Greenberg & Abenavoli, 2017). While we do not yet have sufficient evidence on whether universal SEL interventions are equally effective, more beneficial, or potentially harmful for vulnerable groups, such as refugees and girls in Diffa, emerging research suggests that considering the needs of these marginalized children in the design of SEL programs appears to yield greater benefits for them compared to their peers (Deitz et al., 2021; Kim et al., 2024).

Implementing SEL programs in crisis-affected settings like Diffa faces substantial challenges, including resource constraints, an influx of refugees and IDPs, and struggles to meet students' diverse needs. In such contexts, it is difficult to implement an SEL intervention within the public school system and obtain the necessary support and commitment from schools that such interventions often require. In addition, due to the various challenges and competing priorities, many children's attendance in any program is likely infrequent (Aber et al., 2021; Brown et al., 2023; Roby et al., 2016). This lack of continuity poses a barrier to implementing sequenced lessons and activities, limiting the exposure and dosage necessary for SEL interventions to make meaningful changes in children's skill development. Lastly, while the need for social and emotional support is great for children in crisis-affected settings, the unpredictability of the field conditions and limited time window for funding reduces the probability of long-term implementation. Given the temporary nature of humanitarian programs, and the challenges of recruiting well-trained educators in conflict-affected low-resource settings, it is also difficult to provide adequate support for teachers for an intervention that requires extensive support and training (Dickson & Bangpan, 2018).

Given these implementation challenges, SEL interventions in conflict- and crisis-affected settings need to take field-feasible approaches. *Skill-targeted SEL activities* are a type of SEL intervention consisting of sets of short, targeted strategies or activities that focus on improving specific social and emotional skills for children. This approach makes the intervention more easily deployable and adaptable in the field, requires minimal training, and depends less on the strict sequence and structure of the program components to elicit the intended treatment effect than comprehensive interventions, potentially increasing uptake, impact, and sustainability (Jones et al., 2017). First, *Skill-targeted SEL activities* can be flexibly deployed and implemented for short periods, multiple times a day, within existing programming such as formal or informal academic education programs that have limited school/program days and short hours. Skill-targeted SEL activities are sets of standardized, bite-sized activities that do not require specific sequencing. Thus, they also provide flexibility and freedom of choice for implementors to select among the "menu" of available activities, adequate for conflict-affected low-resource settings where consistent attendance and implementation continuity—and therefore fidelity—cannot be guaranteed. Lastly, the training and support needs for the facilitators to implement SEL activities are likely to be more targeted and lighter, compared to more comprehensive SEL interventions that require more substantial content knowledge and/or application in general classroom instructional practices.

Several SEL interventions using this type of delivery approach are beginning to emerge and be implemented in crisis-affected, low- and middle-income countries (LMICs) with variable results in its impacts on both academic and social and emotional outcomes thus far (Brown et al., 2023; Shah, 2017; Tubbs Dolan et al., 2022). As the design, implementation, and evaluation of skill-targeted SEL activities are still developing, there is a need for further evidence on their feasibility, effectiveness, adaptability, and sustainability to assess its viability, especially in crisis-affected LMICs.

Skill-Targeted SEL Activities: Mindfulness and Brain Games

This study evaluates the effectiveness of the sequential implementation of two skill-targeted SEL activities: Mindfulness and Brain Games. These activities focus on strengthening specific social and emotional skills—i.e., stress management, emotion regulation, executive function–that are likely to be affected by exposure to prolonged and severe adversities such as those that occur during or as a result of poverty, conflict and forced migration (Buckner & Kim, 2012; Keresteš, 2006; Khamis, 2019; Qouta et al., 2008; Shonkoff et al., 2012) and critical for improving learning and adaptation (Eisenberg et al., 2005; Finch et al., 2022; Kim et al., 2020). Mindfulness and Brain Games activities were adapted and implemented by an international non-governmental organization (INGO), and designed to supplement the existing after-school remedial education program called Tutoring in a Healing Classrooms (HCT). HCT is one of the few education programming with evidence of effectiveness in conflict- and crisis-affected contexts, and currently being widely implemented across many of the over 40 countries the INGO provides education services in.

Mindfulness and Brain Games directly target skills that can be compromised in conflict-affected environments. The decision to provide and evaluate these interventions in Niger was informed by theoretical justification and evidence from high-income, stable contexts, where similar activities have demonstrated positive impacts on children's stress responses, emotional regulation, and behavioral outcomes. Recognizing the lack of causal evidence in conflict-affected, low-resource settings, this study aimed to address a critical gap by rigorously evaluating the feasibility and effectiveness of these interventions in a context marked by significant adversity.

Mindfulness activities are designed to improve emotion regulation and stress management, which are critical given the ongoing exposure to trauma and chronic stress experienced by children in crisis and conflict-affected contexts (Eruyar et al., 2018; Khamis, 2019; Reed et al., 2012). These practices are relatively simple to implement and can offer children immediate techniques to down-regulate stress responses, an essential skill for coping with instability. Brain Games, on the other hand, focus on strengthening executive function skills such as working memory and inhibitory control, which are foundational for learning and adapting in unpredictable settings. Children in Niger's conflict-affected areas often experience cognitive disruptions due to trauma and the constant threat of violence; this can make it difficult to concentrate, follow instructions, or engage in classroom learning (Shonkoff et al., 2012). Brain Games provides structured opportunities to practice these skills in a playful, engaging way. Both of these skill-targeted activities are short, flexible activities that do not require extensive training or infrastructure, making them suitable for implementation in schools where teacher capacity, attendance, and classroom stability are often inconsistent.

HCT is an academic tutoring program built on "*climate-targeted SEL*" approach, which focuses on providing a safe and supportive classroom climate supportive of social and emotional development through incorporating positive classroom practices, without providing explicit SEL instruction. HCT, like other "*climate-targeted SEL*" programs in US and high-income country contexts, does not directly target specific SEL skills and evidence suggest this approach has little to no impacts on SEL outcomes (Cipriano et al., 2023; Deitz et al., 2021; Kim et al., 2024). Prior trials of HCT demonstrated positive impacts on academic outcomes and perceptions of the safety and supportiveness of school environments in Democratic Republic of Congo, Lebanon, Nigeria, and Niger (Brown et al., 2023; Diazgranados Ferráns et al., 2022; Torrente et al., 2019; Tubbs Dolan et al., 2022), but no impacts on children's SEL skills (Torrente et al. 2019; Tubbs Dolan et al., 2022).

Mindfulness consists of brief mindfulness activities, such as breathing techniques and mindful movement, developed based on existing mindfulness practices (Greenberg & Harris, 2012) and the activities from interventions such as the *MindUp* Program (Maloney et al., 2016). Focusing on mind-body regulation, these activities are designed to help children manage stress and regulate emotional responses and behaviors in the face of stressful or emotionally-arousing events in the short term (proximal outcomes), and in the longer term, to improve general cognitive regulation (executive function), social and emotional functioning, and academic outcomes (distal outcomes: see Figure 1 for the program theory of change). A growing body of literature supports the benefits of mindfulness-based interventions in improving stress responses, executive function, attention, behavioral regulation, social skills, and mental health (Dunning et al., 2019). However, evidence is more mixed when only rigorous evaluations with randomized controlled trials were considered, with significant effects on proximal outcomes (stress, mindfulness/emotion regulation) but not on distal/secondary outcomes (e.g., executive function). These mixed findings are likely due to the variability in program fidelity and dosage, subgroup difference in baseline and malleability (Dunning et al., 2019; Kuyken et al., 2022). In conflict-affected settings, similar Mindfulness interventions have shown initial promising results with children in Jordan and Palestine (Shah, 2017) and post-conflict Northern Uganda (Matsuba et al., 2021).

Brain Games consist of short and simple games that use movement and playfulness to practice executive function skills (attention, working memory, inhibitory control) necessary for learning (proximal outcomes: Barnes et al., 2021). Through the improvement of these skills, children are better able to regulate their behavior and emotions in classrooms, leading to better academic gains (distal outcomes: Finch et al., 2022; Kim et al., 2020). A recent impact evaluation study of Brain Games implementation in the U.S. yielded marginal positive effects on behavioral regulation, attention control, and impulsivity (Barnes et al., 2021).

In Niger, Brain Games were designed to be implemented after the Mindfulness. This sequential implementation was based on a hypothesis that emotion and stress regulation skills, target outcomes of the Mindfulness activities, are the necessary

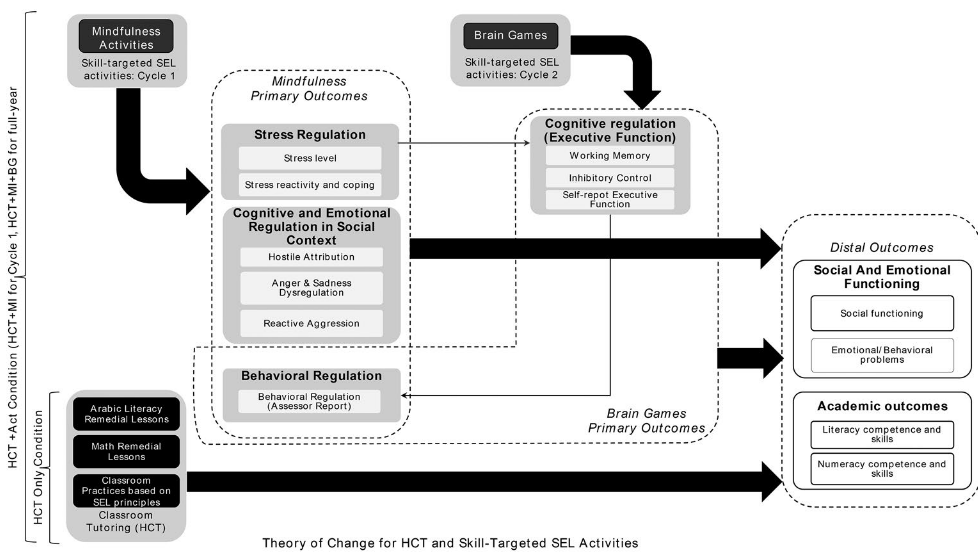

Figure 1. Theory of change for mindfulness and brain games sequential implementation, added to learning in healing classroom tutoring programming.

conditions for developing cognitive regulation skills, such as working memory and inhibition (Jones et al., 2017; Figure 1). The decision to test two SEL interventions sequentially are made to balance 1) maximizing the evidence generation to inform program design and implementation decision-making for the implementing organization within a short time frame afforded given the programming and research funding; and 2) responding to the call for evidence on the sequence and combination of SEL skills taught (e.g., emotion skills preceding cognitive skills) relate to the effectiveness of SEL programming (Cipriano et al., 2023; Denham et al., 2012; Izard et al., 2001).

Similar interventions were concurrently implemented and evaluated in Lebanon with Syrian refugee children attending Lebanese public schools (Tubbs Dolan et al., 2022) and in Sierra Leone in regions affected by the Ebola crisis (Wu et al., 2022). In Lebanon, the combination of Mindfulness activities with the HCT had negative impacts on school-related stress and stress reactivity, while having marginally positive impacts on emotional and behavioral regulation skills when compared to a public school-only condition (Tubbs Dolan et al., 2022). However, neither the 16-week Mindfulness activities implementation (Tubbs Dolan et al., 2022) nor the full-year of skill-targeted SEL activities implementation—which consisted of 16-week Mindfulness followed by 16-week Brain Games—demonstrated unique impacts on the tested SEL outcomes, over and above the HCT (Brown et al., 2023). In Sierra Leone, Mindfulness and Brain Games activities were implemented simultaneously in public school settings, alternating activities from two programs each session of the day. An implementation study found that higher dosage, content coverage, and duration of SEL activities implementation were positively associated with students' school attendance rates, decreased concentration problems, and increased prosocial behavior (Wu et al., 2022).

While the findings from the concurrent studies in Lebanon and Sierra Leone did not directly inform the decision to implement Mindfulness and Brain Games interventions in Niger, they highlight the importance of building a cumulative evidence base to understand how SEL interventions function across diverse populations and contexts. This study in Niger is one of the first large-scale, cluster randomized controlled trials to evaluate skill-targeted SEL activities in a conflict-affected, low-income country. By addressing a significant gap in the global evidence base, it contributes to the growing body of research on the applicability and potential effectiveness of SEL programming in conflict-affected settings.

Current Study

This study examines the effectiveness of the two sets of skill-targeted SEL activities (Mindfulness: MI, Brain Games: BG) embedded within a non-formal remedial education program (Healing Classrooms Tutoring: HCT) on the learning and wellbeing of the Nigerian refugee and Nigerien children in Diffa region of Niger, affected by Boko Haram attacks. Specifically, we aim to evaluate the intent-to-treat impacts of (1) Mindfulness activities implemented for the first 11 weeks of the programming (Cycle 1: HCT+MI) compared to 11-week HCT-only programming; and (2) sequential implementation of Mindfulness and Brain Games after the full-year of implementation (22 weeks, Cycle 1 and 2: HCT+MI+BG) compared to 22-week HCT-only programming, on proximal outcomes including children's social and emotional skills that Mindfulness and Brain Games directly targeted, as well as practice- (social and emotional functioning in classrooms) and policy-relevant distal outcomes (academic outcomes), based on the existing theory and evidence (Cipriano et al., 2023; Durlak et al., 2011; Dunning et al., 2019). See Online Supplement A for more details on implementation design and treatment contrasts. In addition, we explore (3) the variation in the impacts by gender and refugee status on children's outcomes.

The current study is one of the first, relatively large-scale, cluster-randomized controlled trials of skill-targeted SEL programming in the contexts of conflict and crisis, along with the concurrent trial evaluated in Lebanon (Brown et al., 2023; Tubbs Dolan et al., 2022). It builds on a previously-reported trial that evaluated the impact of HCT programming in the Democratic Republic of Congo (Torrente et al., 2019), as well as the added impacts of sequential 22-week implementation of the two skill-targeted SEL activities (Mindfulness, Brain Games) on academic outcomes collected as a part of program monitoring and administrative data with a larger sample in Niger (Brown et al., 2023). In this previous report, the HCT program significantly improved children's French literacy and numeracy, but not in school grades assigned by teachers – an academic outcome that tends to reflect not only children's academic learning but also students' behavior and functioning in schools which reflect students' social and emotional skills (Brown et al., 2023). The addition of skill-targeted SEL activities for the full year—i.e., combined impacts of Mindfulness (first 11 weeks) and Brain Games (second 11 weeks)—to the same HCT program, however, demonstrated positive impacts on school grade averages (Brown et al., 2023), indicating potential positive impacts of the skill-targeted SEL activities on social and emotional outcomes.

Methods

Sampling Procedure and Participants

Figure 2 presents the sampling procedure and overall study design. This study was conducted in 30 public schools in the Diffa department of Niger, a region that shares a border with Nigeria. The schools were selected from a list of 75 potential schools using the following feasibility and safety criteria: security clearance, distance from NGO Office <40km, sufficient number of teachers and students (min. eight teachers and 120 students), serving at least three primary grades. Prior to school selection, local NGO staff visited each school to assess the school eligibility and collect data on school characteristics.

Of the selected 30 schools that met all inclusion criteria (18 in Diffa town, 12 in Maine-Soroa town), 20 were traditional French-only schools, and 10 were French-Arabic schools. French-Arabic schools had a religious focus, with instructional emphasis on Arabic (in addition to French), to read the Koran. The student composition of the schools varied widely, reflecting the ongoing forced displacement and ethnic/linguistic diversity of the region. Specifically, schools ranged from 10-42% of refugee or internally displaced students, 0-85% Kanuri speakers, 0-48% Hausa speakers, and 2-95% Fulfulde speakers. The majority of second- to fourth-graders in these schools struggled academically, with 75-100% unable to read Grade One level texts in French and 61-98% unable to solve simple subtraction problems in the screening tests.

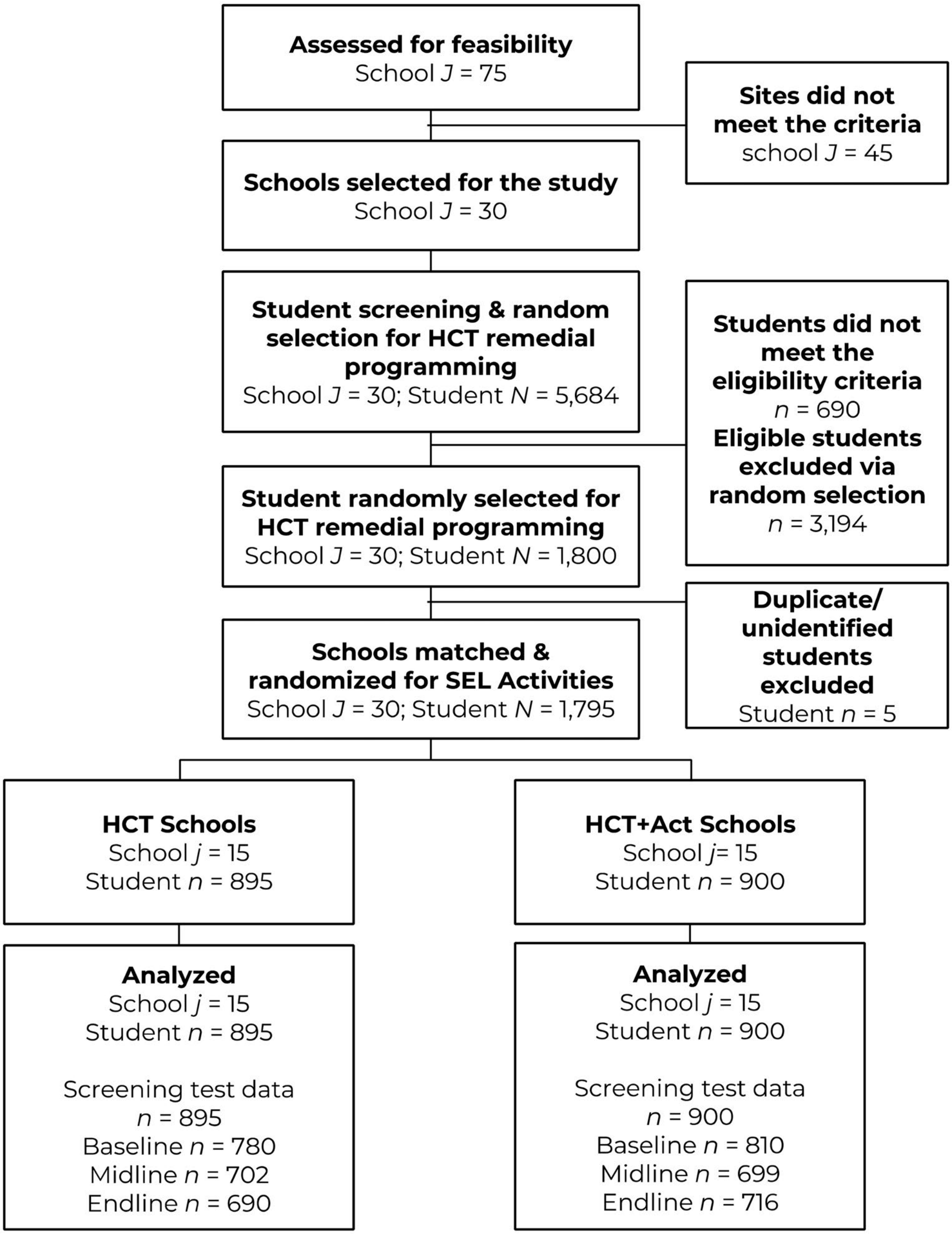

Figure 2. Sample flowchart.

The study included 1,795 second- to fourth-grade children enrolled in the selected schools, with 53% girls and 19% Nigerian refugee children. Participants were chosen from a pool of 5,684 students enrolled in second to fourth grades in the 30 selected schools. Of those, 88% ($N=4,994$) were eligible for tutoring due to low performance in French literacy and mathematics screening tests, the Annual Status of Education Report (ASER: Banerji et al, 2013). However, due to limited resources and funding constraints, IRC could not offer remedial programming to all eligible children. Therefore, to ensure fairness and avoid stigmatization for attending remedial programming, we randomly selected 1,800 from the

4,994 eligible students and offered them participation in the HCT program. These selected students were assigned to 90 after-school HCT classrooms in 30 participating schools. The students who met the eligibility criteria but were not selected via lottery ($n = 3{,}194$) were excluded from the study and were not offered HCT. Of the selected 1,800 children, five were found to be either duplicate entries or unidentifiable due to administrative errors, and therefore excluded from the final sample ($N = 1{,}795$).

Research Design

The current study employed a matched-sample cluster randomization design. This approach can increase power, statistical efficacy, and robustness of causal impact estimates by reducing the heterogeneity within pairs and between treatment conditions (Imai et al., 2009). Schools were matched on a vector of pre-randomization characteristics: school characteristics (e.g., school type, number of classrooms and teachers) and student composition (e.g., refugee status, mother tongue, baseline student math and literacy performance), using the optimal nonbipartite matching (Zhang & Small, 2009) procedure with the nbpMatching package on R (Lu et al., 2011). This matching procedure uses Mahalanobis distance scores between all possible pairs of clusters given all covariates to produce the set of best-matched pairs with the minimum total multidimensional distance within pairs. Within each pair, one school was randomly assigned to the HCT condition and offered the HCT program only ((HCT schools: 22-week HCT programming); and the other offered both HCT and additional skill-targeted activities (HCT+ schools: 22-week HCT programming, with 11-week Mindfulness (Cycle 1) + 11-week Brain Games (Cycle 2)). The characteristics of HCT and HCT+ schools were highly comparable (Online Supplement B). Power analyses[1] accounted for the matched pair school randomization design and nested data structure of students within schools and classrooms, using Optimal Design software (Spybrook et al., 2006). We estimated the minimum detectable effect size of the SEL activities to be 0.17-0.26.

Skill-Targeted SEL Activities Development and Adaptation

Two skill-targeted activities, Mindfulness and Brain Games, were developed and adapted for use in conflict- and crisis-affected settings. Mindfulness activities consisted of 23

[1]Site-level R-squared and intra-class correlation (ICC) estimates for these power calculations are based on: (a) estimates of 15 sub-Saharan African countries provided in Kelcey et al. (2016); and (b) literacy and numeracy impact data from our evaluation of LHC in conflict-affected regions of the Democratic Republic of Congo (DRC: Torrente et al., 2019). School-level intra-class correlation (ICC) estimates in DRC trial (Torrente et al., 2019) was .05, and we assumed similar range of ICCs, .05-.10. In Kelcey et al. (2016), school-level R-squared estimates of socioeconomic status and school location ranged from .02-.59. Given that we matched the schools based on similar variables and also on student baseline performance, we estimated the R-squared explained by the blocking variable to be larger, .50-.70, and a proportion of school-level variance explained by other covariates to be .30-.50. The effect size variability was assumed to be .00-.01, which is realistic upper bound (Raudenbush, Martinez, & Spybrook, 2007). Given effect sizes of mindfulness and executive functioning program evaluations in the US range from 0.25-0.80; and effect sizes in our evaluation of HCT in the DRC ranged from 0.14-0.32, we hypothesized the effect sizes of Mindfulness or Brain Games to be 0.20-0.40. With the matched pair school randomized design of 30 schools, 90 tutoring groups, and an average of 20 students per group, the estimated MDES were 0.17-0.26.

mindfulness practices of three types: (1) *Discovering*, practices on body and emotion awareness; (2) *Experimenting*, attention to different movements and sensory stimulation; and (3) *Accepting*, awareness of the present surroundings, which tended to be more complex activities that required students to be still for longer periods. Teachers were free to choose which activities they implemented each session.

Brain Games (Barnes et al., 2021) activities included short game-like activities designed to help students practice three "brain powers" of executive function: (1) *Focus Power* (flexibility of attention deployment), (2) *Remember Power* (working memory), and (3) *Stop & Think Power* (inhibitory control). Each game was designed to follow a similar structure: explanation of purpose, introduction of the brain power, game instructions with demonstration, game play, and post-game talk. Again, teachers were free to choose which of the 20 Brain Games activities they implemented.

Adaptation workshops were conducted by INGO and partner organization staff trained by the developers of Mindfulness and Brain Games prior to the implementation. These workshops included demonstration and piloting of the activities and reviews by the local staff and teachers. The information gathered during the workshops was used to adapt SEL materials and activities to ensure alignment with the local culture (e.g., gender norms, restrictions around physical movements) and students' experiences (e.g., local names and songs). The activity plans and training manuals were translated from English to French and then reviewed by dual-language staff members for accuracy.

Implementation

Two versions of HCT programming were implemented and evaluated for 22 weeks in school year 2016-2017: HCT and HCT+ programs. Both versions of HCT were segmented as fall and spring cycles: Cycle 1 for 11 weeks from December 2016 to March 2017; and Cycle 2 for 11 weeks from March 2017 to June 2017. The schools assigned to HCT condition implemented HCT throughout the two cycles; and the schools assigned to HCT+ condition implemented HCT and Mindfulness activities for Cycle 1; and HCT and Brain Games activities in Cycle 2.

HCT Implementation

The HCT programs were implemented by 90 IRC-trained public school teachers who provided literacy and Mathematics lessons in a safe and supportive environment three days per week after school, for additional four hours weekly. Tutoring teachers received a five-day training on literacy and numeracy curricula infused with universal SEL principles, (2) ongoing support in the form of monthly peer support groups (Teacher Learning Circles), and (3) aligned one-on-one pedagogical coaching by the regional ministry of education. Teachers were compensated for their training and additional instructional time. The estimated cost of implementing HCT, including the INGO operational cost, basic school support for the whole school (e.g., school kits for teachers, classroom equipment, and furniture for the schools), and program training and implementation, was $263 per student for the full implementation year including the startup period (August 2016-June 2017). See Online Supplement C for the details of the cost estimates.

Sixty-six remedial sessions were intended to be held over the 22 weeks of programming. On average, teachers provided most of the intended dosage for HCT remedial sessions (M = 63.36; SD = 4.25), with 23 teachers reporting an additional day (for a total of 67 days) for make-up classes or orientation (HCT: $M = 63.18$; $SD = 4.65$; HCT+: $M = 63.53$; $SD = 3.86$).

HCT+ Implementation

In addition to the base HCT curriculum, the 45 tutors in the HCT+ schools implemented two skill-targeted SEL activities: Mindfulness and Brain Games. To do so, tutors received an additional three-day training on Mindfulness (prior to Cycle 1) and another three-day training on Brain Games (between Cycles 1 and 2). Tutors in the HCT+ group also received mentoring and support on skill-targeted SEL activities. The cost of adding skill-targeted SEL activities to HCT for the full year was $33, with additional teacher training, materials, and monitoring and evaluation (see Online Supplement C for the details). This program cost is much lower than the typical cash transfer programs' transfer amount in sub-Saharan Africa, $96-$1,344 per household/beneficiary per year (Garcia & Moore, 2012)

In HCT+ schools, teachers were asked to implement Mindfulness (Cycle 1) and Brain Games (Cycle 2) three times per tutoring day: at the beginning, end, and during the subject transition break (total 30 minutes per day, 90 minutes per week, for 33 program days each for Mindfulness and Brain Games). In the HCT condition, these subject transitions were teacher-supervised breaks. Teachers in HCT+ conditions were asked to document the number of activities they implemented each day in weekly activity tracker forms. However, not all teachers submitted weekly forms: 87% of the Mindfulness and 81% of the Brain Games weekly forms were recovered. Of the collected, teachers recorded on average 1.6 activities per day for Mindfulness (53% of the intended dosage); and 1.8 activities per day for Brain Games (60% of the intended dosage).

Student attendance to HCT and HCT+ programs. Students on average attended only about half of the remedial sessions (HCT: $M = 0.50$; $SD = 0.32$; HCT+: $M = 0.48$; $SD = 0.32$). Of the participants, 85% ($n = 1{,}529$) attended at least one tutoring session; the average attendance rate was 66% ($SD = 32\%$), with an average of 77% among those who attended at least one session. Students in HCT+ schools attended HCT sessions at significantly higher rate ($M = 68\%$, $SD = 30\%$) than students in HCT schools ($M = 63\%$, $SD = 33\%$), $t(1793) = -3.68$, $p < .001$.

Data Collection

Data were collected three times across the program year, 2016-2017: December 2016 (Baseline, pre-intervention); Late February to early March 2017 (Midline, at the end of the Cycle 1 program implementation); and May 2017 (Endlines, at the end of the Cycle 2 program implementation). For child assessments, teams of trained data collectors visited HCT program classrooms in the participating schools. Students were paired with the assessors who speak their home language if possible, and if not, speak the same market language they both are fluent in. Data collectors conducted one-on-one tablet-based verbal assessments with each child in a quiet, but not secluded, area.

Each child assessment lasted about 45 minutes to an hour. At the end of each day, data collectors uploaded the data under the supervision of research officers.

Measures

A common set of priority measures (e.g., demographics, literacy) were assessed on the whole sample (core package). To minimize student burden and capture a wide range of children's academic and social and emotional functioning, we assessed children on some measures with a randomly-selected half of the sample (Package A) and another set of measures with the other half (Package B).

Whenever possible, we selected, translated, and adapted measures that had been previously used in similar contexts, e.g., Sub-Saharan African countries and other low- and middle-income country contexts (e.g., see below measures of executive function, behavioral regulation, literacy and numeracy). However, due to the scarcity of locally-developed measures of social and emotional learning in such contexts at the time of this study, in other cases, we adapted and/or assembled from measures developed and tested mainly in the U.S. All measures were translated from English to French, and revised in consultation with local bilingual staff. When possible, performance-based and scenario-based assessments were used in lieu of self-report to avoid bias due to social desirability and developmental differences in metacognition and self-awareness. We conducted cognitive interviews and piloted the majority of the measures prior to the evaluation to assess feasibility, comprehension, cultural/contextual fit, and potential bias. The measures were adjusted based on the feedback and pilot results.

Given this trial was the first time any of these measures were used and tested with children in Niger, we conducted a comprehensive psychometric analysis to establish the evidence validity and reliability of the measures. The psychometric properties of these measures were assessed for factor structure, reliability, and measurement invariance across treatment groups, gender, and refugee status, as well as across three time points, and were found to be acceptable. These psychometric tests were important to establish evidence of the validity and reliability of the measures with this population.

Social and Emotional Outcomes

School-Related Stress Regulation (Package A). Two subscales of the Response to Stress Questionnaire-Academic Problems (RSQ-AP: Connor-Smith et al., 2000) were used: (a) 10-item Academic Problems scale to measure the perceived stress level in school-related academic events (e.g., "doing poorly on a test": α =.93-.94, ω=.95-.96); and (b) 15-item Response to Stress: Involuntary Engagement subscale was used to capture children's involuntary stress reactivity to their perceived school-related stress (e.g., "My thoughts start racing when I am faced with school problems": α =.89-.93, ω=.91-.95). See Kim & Wu (2022a) for evidence of reliability and validity.

Behavioral Regulation (Core Package). Children's behavioral regulation was rated by assessors using the Self-Regulation Assessment-Assessor Report (SRA-AR: Smith-Donald et al., 2007) previously adapted for and used in a study in Zambia (McCoy et al., 2017). We used a shortened 13-item version that had shown strong evidence of reliability and validity in Niger. Assessors rated each child on the behaviors displayed on 13 items (e.g., "Pays attention to instructions and demonstration," "remains in seat appropriately during test") during the child assessments. Internal consistency was high, α = .87-.91 and ω=.92-.96 (Kim & Wu, 2022b).

Executive Function: Performance-Based (Package B). The Rapid Assessment of Cognitive and Emotional Regulation (RACER: Ford et al., 2019) was used to assess working memory and inhibitory control. Working memory was measured using a Spatial Delayed Match to Sample task (Goldman-Rakic, 1996); and inhibitory control was measured using a Simon Task (Simon & Rudell, 1967). RACER has demonstrated good accuracy and reliability in testing with children aged 6-12 in Peru, Lebanon, and Niger (Ford et al., 2019) and has been used in Ghana, Bangladesh, and Ethiopia. See Ford et al. (2019) for task and scoring details and evidence of reliability and validity. Working memory and inhibitory control scores were standardized using baseline data to aid interpretation.

Executive Function: Self-Report (Core Package). At midline and endline, we also collected children's self-report of their own executive function skills using a scale developed for the study (core package), loosely based on a teacher-report measure, Classroom Executive Function Survey (CEFS: Jones et al., 2015). The 10 items ask about children's cognitive and behavioral abilities related to executive function such as working memory (e.g., I am able to remember directions or instructions), attention control (e.g., I am able to pay attention or focus), inhibitory control (e.g., I am able to control my actions), and planning (e.g., I am able to make a plan to complete a task). Internal reliability was acceptable at both time points (α=.82-.86; ω=.88-.91). See Online Supplement D for evidence of reliability and validity.

Cognitive and Emotional Regulation in Social Contexts (Package B). Children's Stories (CS: Dodge et al., 2015) was used to measure children's cognitive and emotional regulation skills in social contexts. CS is a scenario-based assessment of hostile attribution and reactive aggression, and it was previously validated and used in nine countries (Dodge et al., 2015). This study used an adapted version of CS consisting of six hypothetical scenarios of ambiguous peer interactions that could lead to social conflict, followed by questions on their response. In addition to questions on hostile attribution and reactive aggression, we added two questions about emotional dysregulation (sadness, anger) adapted from Di Giunta et al. (2017). This adapted version of CS has demonstrated evidence of validity and reliability (Online Supplement E). Each subscale had acceptable internal reliability (hostile attribution: α=.77-86, ω= .89-.93; anger dysregulation: α=.83-.88, ω=.90-.94; sadness dysregulation: α=.82-.90, ω=.91-.96; and reactive aggression α=.74-.80, ω=.85-.92).

Social and Emotional Functioning: Self-Report (Core Package). We used the child-report version of Strength and Difficulties Questionnaire (SDQ: Goodman, 2001) to capture social and emotional functioning. Original SDQ consists of five subscales (25 items): Hyperactivity, Emotional Symptoms, Conduct Problems, Peer Problems, Prosocial Behavior. Due to inadequate psychometric evidence of these scales, we instead generated and used two new subscales based on exploratory and confirmatory factor analyses: (1) Emotional Problems (13 items, α=.80-.85, ω=.88-.92) and (2) Social Functioning (8 items, α=.74-.81, ω=.83-.90). See Online Supplement F for evidence of validity and reliability.

Academic Outcomes

Literacy Competence. Students' French literacy skills and competence were assessed on the full sample (core package) using the Early Grade Reading Assessment (EGRA: Gove & Wetterberg, 2011) adapted for the current study for use in Niger. EGRA included six subtasks ranging from pre-literacy skills (e.g., receptive vocabulary) to higher-order skills such as reading comprehension. Due to strong floor effects and some ceiling effects, all subtask scores were transformed into proficiency levels with zero scores coded as zero and non-zero scores coded as 1-5 based on percentile ranks (20, 40, 60, 80, 100) based on the baseline distribution. An exception was Reading Comprehension, for which we used raw scores given the small number of items (5). A latent factor consisting of all subtask scores was used to represent children's overall literacy competence. Internal consistency across EGRA subtask scores was high, α=.80-.82; ω=.90-.93.

Numeracy Competence. The Early Grade Mathematics Assessment (EGMA: RTI International, 2014; package A) is an assessment of early numeracy competence, with an emphasis on numbers and operations. The current study used six subtasks ranging from number identification to word problems. Scores for subtasks with more than nine items were transformed using the same quintile transformation used for EGRA. A latent factor identified through factor analysis consisting of all transformed subtask scores was used to represent children's overall numeracy competence. Internal consistency across the recoded EGMA subtasks were high, α=.87-.90; ω=.91-.92.

Covariates

Information on child-level and classroom-level characteristics was used as covariates. From administrative records, we retrieved information on school characteristics, child demographic characteristics, and their screening test scores on French Reading and math (Annual Status of Education Report (ASER): Banerji et al., 2013). In addition, we asked children at baseline on various indicators of socioeconomic status, and other household and child characteristics relevant to the Nigerien local and Nigerian refugee population in Diffa. See Online Supplement G for descriptive information of all covariates used in the impact analyses. School characteristics used in the school matching process were contained within school fixed effects and, therefore, omitted from the models.

Analytic Strategy

All measurement and impact analyses were conducted using Mplus 8.3 (Muthén & Muthén, 1998- 2017), with mean- and variance-adjusted weighted least squares (WLSMV) estimator to address the non-normally distributed or categorical variables.

Nested data structures were accounted for using robust standard errors. Model fits were evaluated using the following criteria: <.05 for the RMSEA and SRMR (Steiger, 2007) and >.95 for the CFI and TLI (Hu & Bentler, 1999).

Models testing treatment impacts were fitted with school-pair fixed effects, accounting for the matched-sample cluster randomization design. The analyses were conducted using a series of structural equation models (SEM) estimated separately for each measure. For example, four outcomes included in Children's Stories were modeled together in a single SEM model (Online Supplement H). The measurement models of each construct at different time points were held scalar-invariant within the SEM models, to estimate baseline-adjusted effects of treatment over and above the longitudinal change of the outcomes. This approach allows the practical interpretation of effect size estimates, placing it on the same scale as the within-individual change estimate over time.

Impacts of Mindfulness activities were tested with the data collected at baseline (pre-intervention) and midline (post-Cycle 1), and the impacts of the Mindfulness + Brain Games sequential implementation were tested with the data collected at baseline (pre-intervention) and endline (post-Cycle 2). In each model, a treatment indicator was included; the HCT+ schools were coded as 1, and the HCT only schools were coded as 0. The coefficient on the HCT+ indicator can be interpreted as the impact of access to Mindfulness (MI: for Cycle 1 impact) or Mindfulness + Brain Games (MI+BG: for full-year impact). Both unadjusted estimates and covariate-adjusted estimates were evaluated. In all models, baseline measurement models were scaled to have a mean of 0 and a variance of 1. Unstandardized impact estimates from these models are reported as effect sizes, which are scaled on the variance of the baseline outcomes of the pooled sample.

To determine the significance of impact estimates, we report both naïve *p* values and *q* values—i.e., *p* values adjusted for Type I error common in multiple hypothesis testing, by controlling the sharpened false discovery rate (FDR: Benjamini et al., 2006). Testing the impact of a single intervention on multiple outcomes increases the risk of an erroneous report of effectiveness simply by chance. However, the FDR adjustment to mitigate this risk is based on assumptions that the true effect could be truly zero (not just "small"), the assumption is unlikely to be true in educational program evaluation. Therefore, strict adherence to the reporting criteria based on adjusted *p* value (*q* value) may elevate the risk of Type II error (Gelman et al., 2012). Given this study is the first evaluation of skill-targeted SEL activities in Niger, we also reported findings with $p<.05$ as promising trends, only to inform future efforts in program design and evaluations.

In addition to the main impact analysis, we explored variations in the impact of the skill-targeted activities by children's refugee status and gender. To do so, we added two interaction terms between the treatment indicator (HCT+) and gender (female) and refugee status to the impact models. Given the exploratory nature of the impact variation analysis, we report and interpret findings that are significant based on *p* values ($p<.05$), unadjusted for multiple comparisons.

Transparency and Openness Statement

This trial is registered with the American Economic Association's registry for randomized controlled trials (AEARCTR-0002125: https://doi.org/10.1257/rct.2125-2.1). All procedures and data use were approved by the the New York University (#IRB-FY2016-1174) and the International Rescue Committee (#EDU 1.00.001) institutional review board. The dataset used for this study has been made publicly accessible at: [Link blinded], and held under a custom license due to the vulnerable status of the subjects. Study materials and analysis codes of this study are available from the corresponding author upon reasonable request.

Results

Attrition and Missing Data

Of the 1,795 children offered access to HCT (HCT schools $n = 895$, HCT+ schools $n = 900$), 1,590 participated in the baseline assessment (HCT schools $n = 780$, 89%; HCT+ schools $n = 810$, 90%), 1,401 participated in the midline assessment at the end of Cycle 1 implementation (HCT schools $n = 702$, 78%; HCT+ schools $n = 699$, 78%) and 1,406 participated in the endline assessment at the end of the Cycle 2 implementation (HCT schools $n = 690$, 77%; HCT+ schools $n = 716$, 80%). The differential missing rates were low (maximum 3%) under conservative assumptions (What Works Clearinghouse, 2020). To account for missing information, all data from the intent-to-treat sample were retained in the analysis, and missing data were addressed through the modeling with WLSMV estimation[2]. WLSMV estimation uses the covariance matrix of all variables and produces consistent estimates under various missing data assumptions, and therefore provides an efficient approach to account for missing data (Asparouhov & Muthén, 2010). See Online Supplement J and K for a series of regression results that (a) demonstrate a lack of evidence of systematic differential missingness by treatment conditions and (b) test whether the covariates are predictive of missingness thus supporting the validity of the WLSMV approach to missingness.

Baseline Equivalency Across Treatment Conditions

The difference in child outcomes between treatment conditions at baseline (HCT+ vs. HCT) were estimated from the SEM models with the latent variables at baseline regressed on the treatment indicator, accounting for the clustered standard error at the classroom level and school-pair fixed effects. The latent variables were standardized at a mean of zero and variance of one, so that the difference estimates are equivalent to effect sizes. We found no significant differences across treatment conditions after FDR adjustment in all outcomes with small difference estimates (ES ≤ 0.25; see Online Supplement L), therefore adequate to generate unbiased impact estimates post-statistical adjustment for baseline measures of outcomes (What Works Clearinghouse, 2020).

[2]We also examined the range of possible effect sizes in the presence of differential missingness using factor scores and without accounting for nested structure and covariates. The resulting Lee bound estimates indicate that our conclusions about program impacts are tenable under extreme assumptions about selection bias (Online Supplement I). However, we do not discuss these results further given the methodological challenges resulting in incomparable estimates of our final results and conceptual difference in interpreting estimates that do not allow estimation of intent-to-treat impacts.

Baseline Difference by Refugee Status and Gender

Online Supplement M presents standardized differences in children's outcomes by refugee status and gender at baseline. When compared to Nigerien children, Nigerian refugee children were at a disadvantage in some outcomes, including lower working memory (b=-0.21, p=.026) and social functioning (b=-0.16, p=.038), as well as higher emotional and behavioral problems ($b = 0.20$, p=.007). Refugees also showed poorer French literacy (b=-0.25, p=.005) and marginally lower numeracy performance (b=-0.17, p=.094). We also found that girls were behind boys in their working memory skills (b=-0.16, p=.016) and in their French literacy competence (b=-0.13, p=.013).

Unadjusted Change in Outcomes in Control (HCT) Condition

The unadjusted change in all study outcomes among children in the HCT group (control group in this study) between baseline (December) to end of Cycle 1 (March), as well as baseline (December) to end of Cycle 2 (May), are useful as a reference for interpreting the direction and magnitude of impact estimates (Online Supplement N). Overall, children in the HCT schools showed significant changes in a positive direction from December to March, including a decrease in hostile attribution, and increases in behavioral regulation, basic numeracy competence, and basic French literacy competence. From December to May, however, these earlier gains disappeared. Exceptions are continued increases in basic numeracy competence and significant reductions in self-reported emotional and behavioral problems. The impact estimates of the skill-targeted SEL activities presented below are over and above these changes in the HCT group.

Aim 1: Intent-to-Treat Impacts of Mindfulness Activities (Cycle 1: HCT+MI)

We report the impacts of access to Mindfulness activities among children who were enrolled in the HCT classes, organized by proximal (targets of Mindfulness activities) and distal outcomes, measured after the Cycle 1 implementation of the program. Both unadjusted and adjusted impact estimates are presented in Table 1; and we report adjusted impact estimates in the text below.

Impacts on Proximal Outcomes

Of the proximal outcomes, we found that access to Mindfulness activities had a significant and positive impact on children's sadness dysregulation skills (Effect size (ES) = -0.30, p = .003; q = .022) and marginally significant positive impacts on reactive aggression (ES = -0.29, p = .029, q = .096) after adjusting for the FDR. That is, among the children who were enrolled in public schools and offered HCT, those who had access to Mindfulness reported steeper decreases in dysregulated expression of sadness and aggressive reactions in social conflict situations, indicating improvement in their capacity to regulate emotions and behavior in social situations.

Among other targeted variables of Mindfulness activities, we found no impacts on stress outcomes (school-related stress), other regulation skills in social contexts (hostile

Table 1. Unadjusted and Adjusted Estimates of the Impacts of adding Mindfulness to HCT (HCT+MI), compared to HCT only, Cycle 1 impact after 11 weeks.

	Unadjusted Impact Estimates				Adjusted Impact Estimates			
	Effect Size	*SE*	*p*	*q*	Effect Size	*SE*	*p*	*q*
Proximal outcomes								
Stress Regulation (Self-Report)								
School-related stress	−0.10	0.10	0.362	1.000	−0.08	0.11	0.459	1.000
Stress reactivity	−0.03	0.15	0.824	1.000	−0.05	0.16	0.743	1.000
Cognitive and Emotional Regulation in Social Context (Scenario-Based)								
Hostile attribution	−0.02	0.10	0.824	1.000	0.00	0.10	0.984	1.000
Anger dysregulation	−0.06	0.14	0.669	1.000	−0.05	0.15	0.751	1.000
Sadness dysregulation	−0.26	0.10	0.006	0.044	−0.30	0.10	0.003	0.022
Reactive Aggression	−0.29	0.14	0.036	0.122	−0.29	0.13	0.029	0.096
Behavioral regulation (Assessor-Report)	0.01	0.09	0.918	1.000	0.02	0.09	0.820	1.000
Distal outcomes								
Executive Function (Performance-Based)								
Inhibitory control	−0.13	0.08	0.105	0.909	−0.11	0.07	0.117	0.704
Working memory	0.06	0.05	0.238	0.909	0.06	0.05	0.177	0.704
Executive Function (Self-Report)[a]		N/A			−0.01	0.04	0.757	1.000
Social and Emotional Functioning (Self-Report)								
Social functioning	−0.03	0.12	0.812	0.909	−0.19	2.20	0.931	1.000
Emotion/behavioral problems	−0.03	0.10	0.756	0.909	−0.10	1.24	0.934	1.000
Academic Outcomes (Performance-Based)								
Basic Numeracy Competence	0.08	0.07	0.209	0.909	0.09	0.07	0.174	0.704
Basic Literacy Competence	0.01	0.10	0.901	0.909	0.00	0.09	0.976	1.000

Note. q values are *p* values adjusted for Type I error common in multiple hypothesis testing, by controlling the sharpened false discovery rate (FDR: Benjamini et al., 2006). Effect sizes were estimated in SEM models where outcome measures are scalar invariant across baseline and post-intervention time point, with baseline outcome factor scaled at mean of 0 and variance of 1. School-pair fixed effects were included in all models; and student-level covariates were included in the adjusted models.

[a]Effect size for the executive function (self-report) are obtained from the standardized estimates because of the missing baseline data for the measure other effect size estimates are standardized on. The standardized estimate for unadjusted model for this outcome is not available due to convergence issue.

attribution, anger dysregulation), or children's behavioral regulation skills during child assessments.

Impacts on Distal Outcomes

Of other outcomes tested, such as executive function, self-reported social and emotional functioning, and academic skills, we found no intent-to-treat impacts of Mindfulness activities.

Aim 2: Intent-to-Treat Impacts of Mindfulness + Brain Games (Cycle 1 + 2: HCT+MI+BG)

We also tested the intent-to-treat impacts of the sequential implementation of Mindfulness (Cycle 1) and Brain Games (Cycle 2) activities on the outcome measured at the end of the program (May) of the children who were enrolled in the HCT classes in HCT+ schools and HCT-only schools. Both unadjusted and adjusted impact estimates are presented in Table 2.

We found that access to the sequential implementation of two skill-targeted SEL activities over the course of the two cycles (one academic year) had no overall impacts on any of the proximal outcomes, neither targeted outcomes of Mindfulness (stress

Table 2. Unadjusted and Adjusted Estimates of the Impacts of adding Mindfulness and Brain Games (HCT+MI+BG) to HCT only condition, after 22 weeks of implementation (full-year implementation of skill-targeted SEL activities).

	Unadjusted Impact Estimates				Adjusted Impact Estimates			
	Effect Size	*SE*	*p*	*q*	Effect Size	*SE*	*p*	*q*
Proximal outcomes								
Stress Regulation (Self-Report)								
School-related stress	−0.01	0.11	0.895	1.000	−0.01	0.10	0.892	1.000
Stress reactivity	0.01	0.16	0.950	1.000	0.02	0.16	0.925	1.000
Cognitive and Emotional Regulation in Social Context (Scenario-Based)								
Hostile attribution	−0.14	1.54	0.93	1.000	0.02	0.13	0.882	1.000
Anger dysregulation	−0.18	2.13	0.931	1.000	−0.08	0.18	0.649	1.000
Sadness dysregulation	−0.70	7.33	0.924	1.000	−0.15	0.13	0.262	1.000
Reactive aggression	0.11	1.39	0.936	1.000	0.01	0.15	0.956	1.000
Behavioral regulation (Assessor-Report)	0.13	3.15	0.967	1.000	0.26	9.52	0.978	1.000
Executive Function (Performance-Based)								
Inhibitory control	0.04	0.06	0.572	1.000	0.04	0.06	0.490	1.000
Working memory	0.00	0.04	0.959	1.000	0.01	0.04	0.797	1.000
Executive function (Self-Report)	−0.10	0.08	0.175	1.000	−0.10	0.89	0.912	1.000
Distal Outcomes								
Social and Emotional Functioning (Self-Report)								
Social functioning	0.00	0.10	0.997	1.000	0.04	2.34	0.988	1.000
Emotion/behavioral problems	0.03	0.10	0.741	1.000	0.57	23.87	0.981	1.000
Academic Outcomes (Performance-Based)								
Basic Numeracy Competence	0.00	0.10	0.976	1.000	0.10	0.07	0.164	1.000
Basic Literacy Competence	0.23	3.47	0.948	1.000	0.02	0.08	0.762	1.000

Note. *q* values are *p* values adjusted for Type I error common in multiple hypothesis testing, by controlling the sharpened false discovery rate (FDR: Benjamini et al., 2006). Effect sizes were estimated in SEM models where outcome measures are scalar invariant across baseline and post-intervention time point, with baseline outcome factor scaled at mean of 0 and variance of 1. School-pair fixed effects were included in all models; and student-level covariates were included in the adjusted models.

outcomes, cognitive and emotional regulation in social context, behavioral regulation) nor of Brain Games (behavioral regulation, executive function) at the end of the school year. We also found no impacts on distal outcomes, which included children's mental health, social and emotional functioning, and academic outcomes.

Aim 3: Impact Variation by Gender and Refugee Status

Table 3 and Figure 3 present the impact variations of HCT+MI, as compared to HCT only, by refugee status and gender. Of 28 interaction terms across 14 outcomes, we found four significant variations. When disaggregated by refugee status, Nigerian refugee children performed better in cognitive inhibitory control tasks ($ES = 0.46$, $p = .017$) than Nigerien children if they had access to Mindfulness activities. We also found that girls benefited more from Mindfulness activities than boys in three out of seven proximal outcomes of Mindfulness. Specifically, girls reported steeper declines in school-related stress ($ES = -0.31$, $p = .020$), sadness dysregulation ($ES = -.31$, $p = .002$), and reactive aggression ($ES = -0.47$, $p < .001$). We found no impact variation for distal outcomes.

We found only one significant variation in the impacts of the HCT+MI+BG full-year implementation (Online Supplement O). Specifically, girls benefited more from the

Table 3. Variation in Impacts of HCT+MI by Refugee Status and Gender.

	(1) School-related Stress			(2) Stress Reactivity								
	ES	*SE*	*p*	*ES*	*SE*	*p*						
HCT+Mindfulness (TX)	−0.08	0.14	0.584	−0.08	0.18	0.663						
TX * Refugee	0.65	0.34	0.053	0.09	0.33	0.779						
TX * Female	−0.31	0.13	0.020	−0.12	0.19	0.545						
Refugee	−0.29	0.19	0.122	−0.01	0.22	0.962						
Female	−0.01	0.11	0.913	−0.10	0.14	0.492						
	(3) Hostile Attribution			(4) Anger Dysregulation			(5) Sadness Dysregulation			(6) Aggression		
	ES	*SE*	*p*	*ES*	*SE*	*p*	*ES*	*SE*	*p*	*ES*	*SE*	*p*
HCT+Mindfulness (TX)	0.05	0.16	0.771	−0.12	0.20	0.528	−0.27	0.15	0.071	−0.44	0.18	0.017
TX * Refugee	0.12	0.23	0.599	0.17	0.25	0.510	0.30	0.29	0.298	0.53	0.42	0.207
TX * Female	0.05	0.12	0.642	−0.26	0.14	0.052	−0.31	0.10	0.002	−0.47	0.13	0.000
Refugee	−0.11	0.16	0.486	−0.22	0.16	0.165	−0.21	0.15	0.172	−0.35	0.21	0.102
Female	0.24	0.14	0.088	0.01	0.15	0.926	0.14	0.13	0.281	−0.02	0.18	0.927
	(7) Behavioral Regulation			(8) Inhibitory Control			(9) Working Memory			(10) Executive Function (self-report)		
	ES	*SE*	*p*	*ES*	*SE*	*p*	*ES*	*SE*	*p*	*ES*	*SE*	*p*
HCT+Mindfulness (TX)	0.20	0.62	0.749	−0.11	0.10	0.254	−0.01	0.07	0.863	0.09	0.59	0.883
TX * Refugee	0.09	0.67	0.895	0.46	0.19	0.017	0.07	0.13	0.626	0.05	0.14	0.743
TX * Female	−0.36	0.91	0.693	−0.16	0.14	0.257	0.11	0.09	0.227	−0.12	0.54	0.820
Refugee	−0.12	0.38	0.746	−0.35	0.15	0.019	−0.09	0.10	0.360	−0.01	0.12	0.962
Female	0.07	0.09	0.483	0.04	0.11	0.725	−0.12	0.07	0.067	0.04	0.05	0.419
	(11) Social Functioning (self-report)			(12) Behavioral/Emotional Problems (self-report)			(13) Basic Numeracy Competence			(14) Basic Literacy Competence		
	ES	*SE*	*p*	*ES*	*SE*	*p*	*ES*	*SE*	*p*	*ES*	*SE*	*p*
HCT+Mindfulness (TX)	−0.07	1.03	0.943	0.12	1.24	0.926	−0.13	0.51	0.802	0.17	0.54	0.756
TX * Refugee	0.10	0.96	0.918	−0.01	1.16	0.995	0.73	0.66	0.274	0.06	0.48	0.908
TX * Female	−0.01	0.81	0.986	−0.19	1.15	0.870	0.18	0.58	0.753	−0.26	0.77	0.736
Refugee	0.02	0.51	0.963	−0.03	0.60	0.962	−0.27	0.44	0.531	−0.23	0.11	0.030
Female	0.10	0.08	0.197	−0.12	0.07	0.073	−0.13	0.09	0.143	−0.11	0.06	0.084

Note. Models (1)-(7) are the models for the proximal outcomes of the Mindfulness activities; and the models (8)-(14) are the models for the distal outcomes of the Mindfulness activities. *q* values are *p* values adjusted for Type I error common in multiple hypothesis testing, by controlling the sharpened false discovery rate (FDR: Benjamini et al., 2006). Effect sizes were estimated in SEM models where outcome measures are scalar invariant across baseline and post-intervention time point, with baseline outcome factor scaled at mean of 0 and variance of 1. Effect sizes reported here are from covariate-adjusted models.

skill-targeted SEL activities in their improvement in working memory performance ($ES = 0.22$, $p = .014$) than boys did at the end of the full-year implementation of the HCT. We did not find any variation of impacts by gender in other outcomes, nor variations of impacts by refugee status.

Discussion

Universal, school-based SEL interventions can be a particularly attractive option to promote healthy development and learning for all children in low-resource contexts

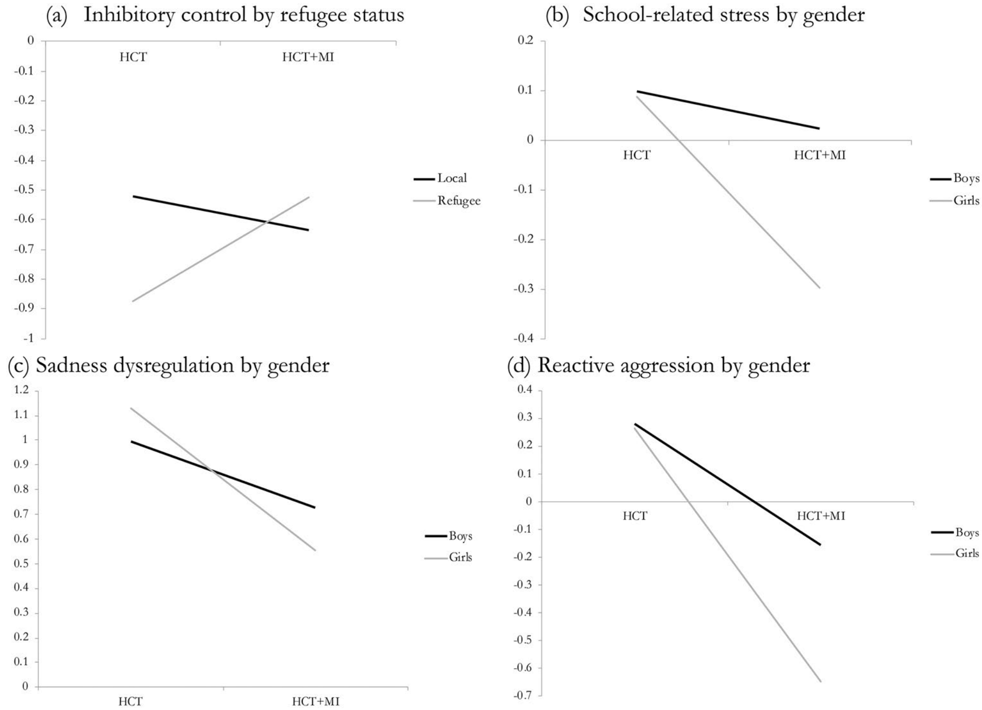

Figure 3. Variation of the HCT+Mindfulness (HCT+MI) Impacts by (a) Refugee Status and (b)-(d) Gender.

affected by conflict and adversity like Niger, given its potential to cost-effectively promote social and emotional skills that bolster resilience (Betancourt et al., 2013; Jordans et al., 2016). This study evaluated the impact of two sets of skills-targeted SEL activities—Mindfulness and Brain Games. These skill-targeted SEL activities are deemed suitable for implementation in conflict- and crisis-affected contexts at a relatively low cost ($33 per student for the full implementation year), when added to supplement the existing after-school remedial program that incorporated a climate-targeted SEL approach (HCT).

This study provides initial evidence that such skill-targeted SEL activities can improve some targeted outcomes in the low-resource, conflict-affected context of Diffa, Niger, when embedded within a supportive learning environment as provided by programs like HCT. As one of the few large-scale trials of SEL interventions in conflict-affected low-resource settings, the initial impacts of 11 weeks of Mindfulness activities show promise of a low-cost, easy-to-implement skill-targeted SEL activities for improving select, but not all, target SEL outcomes among children affected by conflict and crises. In addition, we also found a trend indicating that Mindfulness activities benefit children who are considered to be more vulnerable (girls, refugees) than their peers on some target outcomes related to cognitive, emotional, and stress regulation skills. However, these promising initial impacts of the Mindfulness activities were not sustained when Mindfulness was stopped and replaced with 11 weeks of Brain Games, delivered using a similar format but with different content and targeted skills (executive function). Below, we explore these findings and their implications for future SEL interventions and research in conflict-affected low-resource settings.

The Promise of Skill-Targeted Mindfulness Activities, Especially for Refugees and Girls

After the first 11 weeks of implementation, we found robust positive intent-to-treat impacts of Mindfulness activities—when added to the HCT—that help children in Diffa public schools to better regulate the expression of sad emotions in social conflict situations. We also found promising indications that Mindfulness may reduce reactive aggressions in social situations. While we did not observe impacts on other outcomes, these positive impacts lend tentative support to the hypothesis that the skill-targeted Mindfulness activities embedded within academic remedial education programming can help build emotion regulation skills and reduce aggression among children affected by conflict and adversity. It is well-established, at least in stable contexts, that emotional regulation skills and aggression are related to later social, emotional, and academic learning and future adaptation and well-being (Dodge et al., 2015; Eisenberg et al., 2005). Given such potential downstream impacts, these positive impacts of Mindfulness activities are encouraging.

The patterns of the impact variation also suggest that the skill-targeted Mindfulness activities can lend an additional boost to some of the targeted social and emotional skills for marginalized populations within the harsh environment of Diffa–refugee children and girls. Specifically, we find Nigerian refugee children showing improvement in cognitive inhibitory control skills if they had access to Mindfulness activities, compared to the local Nigerien children in the same condition who did not show significant change. Mindfulness activities also had greater impacts on girls, compared to boys, with significantly greater reductions in sadness dysregulation and reactive aggression outcomes. In addition, we find a unique impact of Mindfulness on girls' school-related stress levels, with a steeper decline from December to March found only among girls who had access to the Mindfulness activities.

The evidence we found provides tentative support for the potential of SEL activities like Mindfulness to give a much-needed boost for skills to cope with the adversities they face, narrowing the equity gap for girls and refugee children in social and emotional learning. While we do not have sufficient information to explain the mechanism of the differential susceptibility by gender and refugee status, similar heterogeneity of impacts has been found in evaluations of mental health and SEL interventions in other conflict-affected context (Jordans et al., 2016; Kim et al., 2024). Given the potential additional benefit or harm for more vulnerable subgroups, SEL programming evaluation in conflict-affected settings must pay special attention to the heterogeneity of impacts in addition to the average effects to guard against potential harm and boost benefits for the most vulnerable.

Null Effects of the Mindfulnesss + Brain Games Sequential Implementation

However, these initial positive impacts of adding 11 weeks of Mindfulness activities to HCT programming dissipated when Brain Games replaced Mindfulness activities for the next 11 weeks. We found no impacts of the sequential implementation of the skill-targeted SEL activities (Mindfulness and Brain Games) after the full-year implementation. While we found the full year of skill-targeted SEL activities improved girls' working memory performance, this finding is difficult to substantiate given the lack of evidence on average outcomes nor on other related outcomes. These results replicate the findings from another trial with a similar design conducted with Syrian refugee children in Lebanon (Kim et al., 2023). These null findings may be due to Mindfulness impact fade-out, the null impacts of the Brain Games, ineffective sequencing, or combination of all these factors.

First, fadeout of Mindfulness activities' impacts may be due to the short implementation period that did not allow sufficient time and support for children to transfer the practices beyond the SEL sessions in schools and incorporate them into their daily routines at home and with friends. The difficulties of engaging children to transfer skills and practices outside of the school-based SEL sessions have been noted as one of the main reasons for the null effects of Mindfulness interventions (Kuyken et al., 2022). Transferring mindfulness practices outside of the school settings may be especially challenging in a context like Niger, where children's safety and wellbeing are frequently threatened, and children may not have access to safe and supportive surroundings to practice mindfulness outside of the classroom. Future mindfulness interventions may consider more sustained SEL activities incorporated into regular school routines and focus on building a supportive environment for practicing mindfulness activities outside of the SEL sessions.

Second, it is possible that the version of the Brain Games implemented was simply not effective in the context of Diffa. It may have been that the Brain Games was not sufficient to show significant impacts over and above the effects of the climate-targeted SEL program (HCT) which was implemented in both conditions in the current treatment contrast. It is also possible the Brain Games activities needed higher quality and quantity of implementation to produce the hypothesized impacts, given its reliance on both repeated practice of EF skills via game plays, some with complicated rules, and post-game talk, which required teachers' capacity to lead meaningful discussion. In a crisis-affected setting like Diffa, there are likely implementation challenges to quantity and variations in dosage, quality, and fidelity, as we have seen in this study with a 50% attendance rate and 53-60% of the intended dosage offered. In addition, security restrictions and curfew reduced the available hours for teacher training, resulting in condensed training days and contents compared to what was originally intended, contributing to reduced dosage and fidelity. Thus, the implementation thresholds to see Brain Games impacts may not have been met.

On the other hand, it is also possible that the Brain Games were not sufficiently adapted to the local values and needs in the culture and context of Niger, compromising its implementation fidelity and quality. In recognition of this possibility, we conducted a post-hoc qualitative study involving teachers, ministry officials, and parents in Niger on their perceptions and acceptability of the SEL programming, after the current SEL intervention implementation was completed (Kabay et al., 2025). The findings indicate a general acceptance and buy-in from teachers and ministry officials. However, parents who were interviewed tended to prioritize moral education and character development over direct support for psychological well-being and academic learning. This insight is invaluable as it highlights the need for ongoing dialogue and adaptation of our programs to better align with local values and expectations. After this trial, there have been increasing efforts to ensure SEL materials are grounded in local values and needs, culturally appropriate, context-relevant, and feasible to implement (Bailey et al., 2021). Such efforts are critical to ensure intervention uptake, impact, and sustainability in conflict-affected low-income countries.

Third, it is also possible that the impacts generated by different activities targeting different skills offset one another in some way or that the activities were not optimally sequenced to produce a cumulative impact. While activities were purposefully sequenced to build skills for physiological down-regulation of emotions prior to addressing more cognitive skills such as working memory and inhibition (Jones et al., 2017), there is little empirical research–even from high-income contexts–to inform whether or how to best sequence SEL skill acquisition (Cipriano et al., 2023). Future research is needed to inform decisions on optimizing of SEL sequencing–both within an intervention and of multiple interventions across the developmental progression over time.

Limitations

We acknowledge several limitations of this study. First, the research design did not allow for testing the unique impacts of Brain Games, nor the effects of differential sequencing of Mindfulness and Brain Games. We made these design decisions to balance the budgetary and programmatic restrictions on the ground, INGO's priority to test both skill-targeted SEL activities within an academic year to inform programmatic decisions for the following years, and limited sample size, power, and research budget. Nonetheless, these decisions made it impossible to estimate the unique impacts of Brain Games nor the impact of the implementation sequencing.

Second, the study design did not include a no-treatment control group, which limits our ability to evaluate the combined effects of skill-targeted SEL activities and the HCT. Although Brown et al. (2023) found significant academic benefits of adding SEL activities to the HCT, compared to a no-treatment condition, this study did not assess SEL outcomes. If we had included a no-treatment condition, we might have observed more pronounced effects of the added SEL activities, given the larger treatment contrast. However, practical constraints, limited research budget, and ethical considerations of repeatedly assessing children without providing corresponding support (Saks et al., 2002), prevented this approach.

Relatedly, some of the null effects reported in this study may be due to limited statistical power. The sampling and research design decisions were informed by the best available evidence at the time, including estimates of ICCs, variances, and effect sizes from similar SEL interventions in Sub-Saharan African contexts (e.g., Kelcey et al., 2016; Torrente et al., 2019). However, research on SEL outcomes remains limited, particularly regarding the malleability, distributional patterns, and variance of key social and emotional skills, which are critical for establishing robust priors for power analysis. These limitations, combined with the absence of no-treatment control conditions and the complexity of the sequential treatment design, likely reduced the study's power to detect effects on some outcomes. Future research should prioritize addressing these gaps by (a) documenting ICCs, variances, and distributional characteristics of SEL outcomes across diverse contexts, and (b) estimating effect sizes of SEL program impacts on various outcomes in underrepresented, low-income, and conflict-affected populations. Such advancements are essential for improving power calculations and research designs in this field.

Lastly, we are unable to isolate the full-year treatment effects from the impact of seasonal timing. The end of the SEL activities implementation coincided with summer, agricultural season, and the end of the school year, which may have resulted in uneven implementation and lower attendance. In anticipation of such seasonal variation, the implementation partner provided school materials as rewards to boost attendance in the last two months of implementation. However, given seasonal variation in teaching and learning (Plank & Condliffe, 2013; Scales et al., 2020) it is questionable how the increased attendance translated to the implementation and uptake.

Contribution and Implication

This study is one of the few large-scale cluster-randomized controlled trials to evaluate the effectiveness of universal school-based SEL interventions on conflict-affected children's social and emotional outcomes in low-income countries. The positive impacts of 11 weeks of the skill-targeted Mindfulness activities, albeit temporary, provide rare evidence of the effectiveness of SEL interventions in conflict-affected, low-income countries based on randomized controlled trials, strengthening support for investing in similar approaches. The additional benefit of the Mindfulness activities with the marginalized refugee children and girls points to promising directions for universal school-based SEL interventions like Mindfulness activities as a low-cost and effective SEL support approach to support all children at scale while closing the equity gap for marginalized children in conflict-affected contexts even in challenging environments like those experienced by children in Diffa, Niger.

Moreover, our findings have broader implications for education in low-resource and conflict-affected settings, where development partners often lead programming innovations. HCT, one of the few climate-targeted SEL programs with proven effectiveness in crisis contexts, is widely implemented by the INGO partner operating in over 40 countries, extending the relevance of this study beyond Niger. The results highlight practical lessons for scaling and adapting SEL interventions in similar settings, particularly for vulnerable populations such as refugee children. By demonstrating how low-cost SEL activities can enhance existing educational supports, this study informs strategic investment and implementation decisions critical for policymakers, program developers, funders, and development partners in conflict-affected contexts.

On the other hand, the lack of sustained impact of Mindfulness and probable null impact of Brain Games suggests that we have a lot of work ahead—to achieve culturally- and contextually-responsive, targeted, sustained, and high-quality program design and implementation, as well as to mount the descriptive and evaluation research needed to provide evidence for such programmatic decisions and practices. Combined with emerging evidence for locally-based and context-responsive SEL intervention design in conflict-affected low-resource settings, this study adds to the small but growing global evidence base for developing, adapting, and improving SEL support for some of the most vulnerable children in the world.

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Author Note

This trial is registered with the American Economic Association's registry for randomized controlled trials (https://doi.org/10.1257/rct.2125-2.1).

All data presented in this study were are available under a creative commons license and can be accessed at: Aber J. L., Brown, L, Dolan, C. T., Kim, H. Y., Annan, J. (2022), *Niger Year 1 Deidentified Data (2016-2017)* [Dataset]. Harvard Dataverse. https://doi.org/10.7910/DVN/TVXIEW

Please contact the corresponding author to inquire about the study materials and analysis code.

Open Research Statements

Study and Analysis Plan Registration

The study and analysis plan are registered at the American Economic Association's registry for randomized controlled trials (RCT ID AEARCTR-0002125): https://www.socialscienceregistry.org/trials/2125/history/94002

Data, Code, and Materials Transparency

The data that support the findings of this study are openly available in Harvard Dataverse: https://doi.org/10.7910/DVN/TVXIEW. The materials and code associated with this study are not publicly available.

Design and Analysis Reporting Guidelines

This manuscript was not accompanied by a completed copy of the JREE Randomized Trial Checklist.

Transparency Declaration

The corresponding author (the manuscript's guarantor) affirms that the manuscript is an honest, accurate, and transparent account of the study being reported; that no important aspects of the study have been omitted; and that any discrepancies from the study as planned (and, if relevant, registered) have been explained.

Replication Statement

This manuscript reports an original study.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Open scholarship

This article has earned the Center for Open Science badges for Open Data and Preregistered. The data and registration are openly accessible at https://doi.org/10.7910/DVN/TVXIEW and https://doi.org/10.1257/rct.2125-2.1.

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